SDUHSD CAREER TECHNICAL EDUCATION

AP Computer Science A

Campus: Canyon Crest Academy
CTE Industry Sector: Information and Communication Technologies
CTE Pathway: Software and Systems Development
Course Sequence: Course 3- Capstone
Prerequisites: AP Computer Science Principles
10 credits, practical art or elective, weighted grade, UC Approved Area G
Recommended Pathway Sequence:

Course 1	Course 2	Course 3	Enrichment
Computer Science Foundations OR Computer Game Design	Human-Computer Interaction (HCI) Design		AP Computer Science Principles AP Computer Science A Machine Learning

Course Description:
AP Computer Science A is equivalent to a first-semester, college-level course in computer science. The course introduces students to computer science with fundamental topics that include problem solving, design strategies and methodologies, organization of data (data structures), approaches to processing data (algorithms), analysis of potential solutions, and the ethical and social implications of computing. The course emphasizes both object-oriented and imperative problem solving and design using Java language. These techniques represent proven approaches for developing solutions that can scale up from small, simple problems to large, complex problems. The AP Computer Science A course curriculum is compatible with many CS1 courses in colleges and universities.

Course Overview (Link to College Board Description of Course and Exam)

Course Syllabi:

CCA

Grading:

See teacher Syllabi

Work load: (High, Medium, Low)

Class setting:
See teacher Syllabi

(This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class).
(Highly independent, Moderately Independent, Structured)

*Note: Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students.

Textbooks or Supplemental Materials for this course:

See teacher Syllabi

Software, program knowledge specific to this course:

Java - Object Oriented Programming using various IDE applications

Industry Connections:

Units of Study, Large Projects, Work-based Learning Connections:

---------- Week 1-2 ----------

Unit 0A – Computer Science, Computer Lab, and Objects Introduction
AP Topics – Test classes and libraries in isolation; Identify and correct errors : compile-time, run-time, logic; Categorize error: compile-time, run-time, logic; Employ techniques such as using a debugger, adding extra output statements, or hand-tracing code; Understand and modify existing code; Inheritance; Object-oriented development; Top-down development; Encapsulation & information hiding;

Student Objectives - Students will learn what Computer Science is, how a computer lab works, how to use the computer, how the network is setup, and how to use the labs and the network in an acceptable/ethical manner. Students will learn the basic syntax for Java and how to debug a program, the difference between a compile error and a syntax error, how to identify and correct errors, how to add to and remove from existing code. Students gain experience working with a large program, modifying existing code of a large program, and expanding and extending existing code.

Unit 0B - Computers
AP Topics – Primary and secondary memory; processors; peripherals; language translators/compilers; virtual machines; operating systems; networks; single-user

systems; networks; system reliability; privacy; legal issues and intellectual property; social and ethical ramifications of computer use.

Student Objectives – Students will learn all of the fundamental components of a computer, how a computer works, hardware, software, compilers, programming languages, basic computer operations, integrity, and responsible use of the computer.

Units A and B – Output and Variables
AP Topics – Primitive types vs. Objects; Constant declarations; Variable declarations; Console output; Java library classes; Simple data types(int, boolean, double); Classes; Representations of numbers in different bases; Limitations of finite representations;

Student Objectives – Students will learn what a variable is, how to define a variable, how to assign values to a variable, the difference between a primitive type and a reference, and how to print/println values to the console window.

Units C and 1 – Input and Methods
AP Topics – Variable declarations; Console output; Java library classes; Simple data types(int, boolean, double); Classes; Method declarations, Class declarations; Parameter declarations.

Student Objectives – Students will learn how to perform basic input operations, write methods, define and pass parameters, and use graphics to make shapes and pictures.

Units 2 and 3 – Classes, OOP, Math Operations
AP Topics – Object Oriented development; Top-down development; Encapsulation and information hiding; Procedural abstraction; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations.

Student Objectives – Students will learn how to declare a class, class methods, and parameters, the difference between constructors, accessors, and modifiers, learn how to read and understand a problem description, purpose, and goals. Students will learn to solve problems using mathematical operators(+,-,/,*,%), mathematical formulas, and Math class methods.

---------- Week 2-3 ----------

Unit 4 - Strings and OOP
AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations; AP CS A Magpie lab introduction and extension.

Student Objectives – Students will learn how to instantiate a String, more about references, how to create a reference to a String, perform String input and output, how to use String methods(length, substring, indexOf, charAt), how to write return methods(toString), and how to create more sophisticated classes.

Units 5-7 – Conditionals – If, If else, If else if, Switch Case
AP Topics – Conditional; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations AP CS A Magpie lab introduction and extension.

Student Objectives – Students learn how to use if, if else, if else if, and switch case to test conditions and add decision making to their programs, and Boolean conditions and variables. Students learn how to use relational operators(>,<,>=,<=,!).

---------- Week 3-5 ----------

Units 8-9 – Iteration – For Loop and While Loop

AP Topics – Iteration; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations.

Student Objectives – Students learn how to use for loops, use while loops, add iterative processes to their programs, and use Boolean conditions and variables. Students learn the different parts of a loop and when to use a particular type of loop. Students will learn when to use Integer.MAX_VALUE and Integer.MIN_VALUE.

Unit 10 - Boolean Logic and Boolean Laws

AP Topics – Boolean; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations.

Student Objectives – Students learn boolean laws, truth tables, logical operators(&&, ||, !, ^), how to use do while loops, how to use boolean logic to solve problems, and how to use Random and Math.random() to generate random numbers.

Unit 11 - Iteration – Nested Loops

AP Topics – Iteration; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations;

Student Objectives – Students learn how to use nested loops, add iterative processes to their programs, and use Boolean conditions and variables. Students learn how to use nested for and nested while loops.

Units 14-15 - One dimensional arrays [!!Critical Topic!!]

AP Topics – One-dimensional arrays; Traversals; Insertions; Deletions; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations.

Student Objectives – Students will learn how to instantiate a one-dimensional array, add items to a one-dimensional array, delete items from a one-dimensional array, and use a one-dimensional array to solve problems. Students will learn the differences between arrays of primitives and arrays of references.

---------- Week 5-7 ----------

Unit 16 - ArrayList [!!Critical Topic!!]

AP Topics – One-dimensional arrays; Traversals; Insertions; Deletions; Object Oriented development; Searching; Sorting; Test classes and libraries in isolation; Identify boundary cases and generate appropriate test data; Perform integration testing; Choose appropriate data representation and algorithms; extensive work with AP CS A Elevens Lab to assist in teaching lists of references.

Student Objectives – Students will learn how to add to, delete from, sort, search, and perform all types of manipulations on an ArrayList. Students will learn about the java.util.List interface. Students gain experience working with a large program, modifying existing code of a large program, and expanding and extending existing code.

Unit 17 – References / Parameters

AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations; AP CS A Elevens Lab modification and extension.

Student Objectives – Students will learn more about references and parameter passing. Students will learn the differences between passing primitives and references as parameters.

Unit 18 - OOP / Array of References / List of References [!!Critical Topic!!]

AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations; One-dimensional arrays; Traversals; Insertions; Deletion; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design;

Student Objectives – Students gain experience working with existing code and modifying, expanding, and extending existing code. Students will learn more about storing references in arrays. Students will learn the difference between arrays of primitives and arrays of references.

---------- Week 8-9 ----------

Unit 19 - Interfaces

AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations; Interface declarations; New AP CS A Labs – Elevens and Magpie

Student Objectives – Students will learn how to design and implement a class; apply data abstraction and encapsulation; and implement an interface and learn why interfaces are useful. Students will learn how interfaces are used to build hierarchies. Students gain experience working with a large program, modifying existing code of a large program, and expanding and extending existing code.

Unit 20 – Inheritance [!!Critical Topic!!]

AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations; Interface declarations; Read and understand class specifications and relationships among the classes("is-a", "has-a"

relationships); Understand and implement a class hierarchy; Identify reusable components from existing code using classes and class libraries; Choose appropriate data representation and algorithms.; Extend a class using inheritance; work with *AP CS A Labs – Magpie and Elevens.*

Student Objectives – Students will learn how to extend a given class using inheritance, design and implement a class hierarchy, write a multi-tiered game with graphics and animation. Students will learn how to build a new class from an existing class using extends and super calls. Students will learn how to use static variables. Students gain experience working with a large program, modifying existing code of a large program, and expanding and extending existing code.

---------- Week 10-12 ----------

Unit 21 - Abstract Classes – BlackJack - Elevens

AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations; Interface declarations; Read and understand class specifications and relationships among the classes("is-a", "has-a" relationships); Understand and implement a class hierarchy; Identify reusable components from existing code using classes and class libraries; Choose appropriate data representation and algorithms; Extend a class using inheritance. Use a List of classes to build a group of objects.

Student Objectives – Students will learn how to design and implement an abstract class, extend an abstract class to make sub classes, and implement an interface. Students will learn to compare and contrast a class, an interface, and an abstract class. Students will learn when to use an interface, when to use an abstract class, and when to use static variables.

Unit 22 – Matrices [!!Critical Topic!!]

AP Topics – One-dimensional arrays; Two-dimensional arrays; Traversals; Insertions; Deletions; Object Oriented development; Class design; Method declarations; Parameter declarations; Class declarations; Top-down development; Encapsulation & information hiding; Introduction and Extended work with AP CS A Lab PictureLab.

Student Objectives – Students will learn how to instantiate a one-dimensional and two-dimensional array, add items to a one-dimensional and two-dimensional array, and delete items from a one-dimensional and two-dimensional array.

Unit 23 - Recursion

AP Topics – Recursion; Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Class declarations; work with *AP CS A Labs – Elevens and PictureLab.*

Student Objectives – Students will learn how to use recursion to solve problems, the benefits of using recursion, when to use recursion, and the negative effects of using recursion. Students gain experience working with a large program, modifying existing code of a large program, and expanding and extending existing code.

---------- Week 13-14 ----------

Unit 24 - Advanced Sorting and Searching / Comparable

AP Topics – One-dimensional arrays; Traversals; Insertions; Deletions; Object Oriented development; Searching; Sequential Search; Binary Search; Sorting; Selection Sort; Insertion Sort; Merge Sort; Test classes and libraries in isolation; Identify boundary cases and generate appropriate test data; Perform integration testing; Choose appropriate data representation and algorithms; Analysis of algorithms; Informal comparisons of running times; Exact calculation of statement execution counts.

Student Objectives – Students will learn to identify all sorting and searching algorithms, code all sorting and searching algorithms, and to select the appropriate sorting and searching algorithm for the appropriate situation. Students will learn where to use a particular sort/search and the benefits of using a particular type of sort/search.

AP Review Time

AP Topics : Arrays, ArrayList, Inheritance, AP CS A Labs – Magpie, Elevens, and PictureLab
Guided Practice : Past year's free response and multiple choice questions
Guided Practice : Slides, AP CS A Labs – Magpie, Elevens, and PictureLab
Readings : Past year's free response and multiple choice questions
Readings : Review book units

Units 12-13 – Chopping Strings and File Input (Optional Topics)

AP Topics – Object Oriented development; Read and understand a problem description, purpose, and goals; Class design; Method declarations; Parameter declarations; Class declarations.

Student Objectives – Students learn how to use Scanner to chop up Strings, to read data from data files, to instantiate Objects using the data extracted from files. Students learn more about constructor overloading and using a single class for multiple purposes.

The following Model Curriculum Standards connected to this course:

Standards for Career Ready Practice
1. Apply appropriate technical skills and academic knowledge. 4. Apply technology to enhance productivity. 5. Utilize critical thinking to make sense of problems and persevere in solving them. 10. Demonstrate creativity and innovation.

Framework Concept	Software and Systems Development Standards
C1.0 Identify and apply the systems development process.	C1.1 Identify the phases of the systems development life cycle, including analysis, design, programming, testing, implementation, maintenance, and improvement. C1.4 Work as a member of, and within the scope and boundaries of, a development project team.

	C1.5 Track development project milestones using the concept of versions. C1.6 Diagram processes using flowcharts and the Unified Modeling Language.
C3.0 Create effective interfaces between humans and technology.	C3.1 Describe and apply the basic process of input, processing, and output. C3.2 Design effective and intuitive interfaces using knowledge of cognitive, physical, and social interactions.
C4.0 Develop software using programming languages.	C4.3 Identify and use different authoring tools and integrated development environments (IDEs). C4.4 Identify and apply data types and encoding. C4.5 Demonstrate awareness of various programming paradigms, including procedural, object oriented, event-driven, and multithreaded programing. C4.6 Use proper programming language syntax C4.7 Use various data structures, arrays, objects, files, and databases. C4.8 Use object oriented programming concepts, properties, methods, and inheritance. C4.9 Create programs using control structures, procedures, functions, parameters, variables, error recovery, and recursion. C4.10 Create and know the comparative advantages of various queue, sorting, and searching algorithms.
C5.0 Test, debug, and improve software development work.	C5.4 Test software and projects. C5.5 Evaluate results against initial requirements.
C6.0 Integrate a variety of media into development projects.	C6.1 Identify the basic design elements necessary to produce effective print, video, audio, and interactive media. C6.6 Integrate media into a full project using appropriate tools. C6.7 Create and/or capture professional-quality media, images, documents, audio, and video clips.